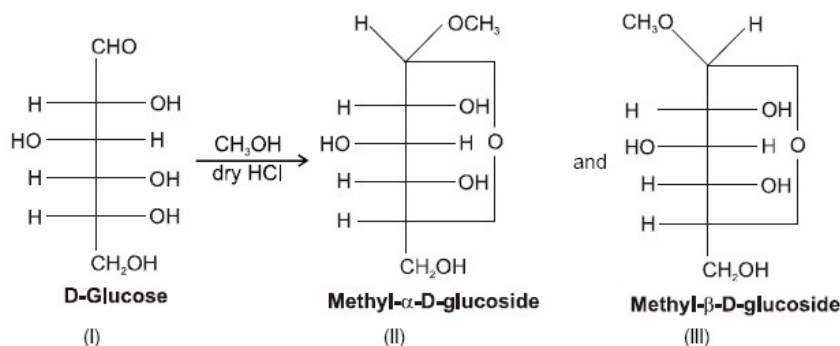


CHEMISTRY

TARGET : JEE- Advanced

**CAPS-25
BIOMOLECULES****SINGLE CHOICE QUESTIONS**

1. D-glucose, on treating with methanol in presence of dry HCl gives methyl glucosides according to the following reaction



Mention true (T) and False (F) from the following statements

S_1 : The glucosides do not reduce fehling's solution

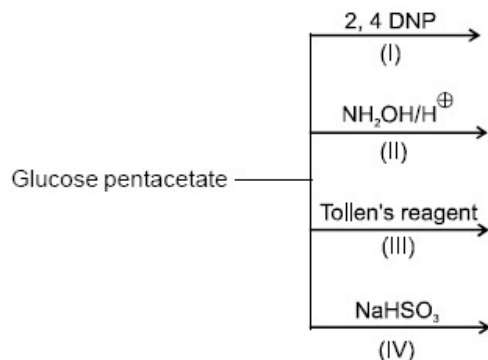
S_2 : The glucosides do not react with hydrogen cyanide or hydroxylamine

S_3 : Behavior of glucosides as stated in S_1 and S_2 indicates the absence of free – CHO group.

S_4 : The two forms of glucosides are enantiomers.

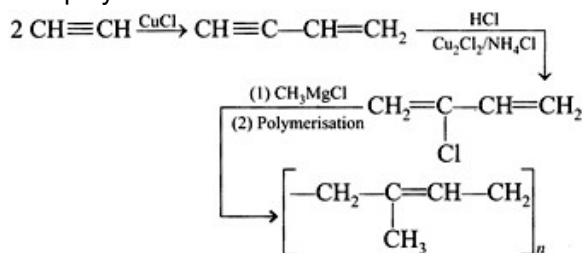
(A) TTFF (B) FTTT (C) TTTF (D) TFTF

2. Observe the following laboratory tests for glucose pentacetate and mention +ve or –ve from the code given below.



(A) + + + + (B) - - - - (C) + - + - (D) + + - -

3. The polymerization reaction shown below



would produce :

(A) PVC (B) neoprene (C) chloroprene (D) Rubber



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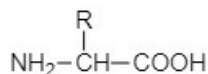
OFFICE ADDRESS : PLOT NUMBER 35, GOPALPURA BYPASS ROAD, NEAR RIDDHI SIDDHI CIRCLE,
10-B SCHEME, TRIVENI NAGAR, GOPAL PURA MODE, JAIPUR, RAJASTHAN 302020

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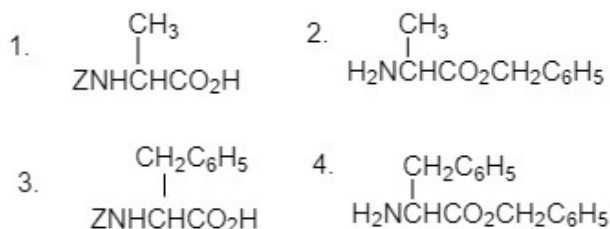
Page # 149

4. An amino acid may be represented by general formula

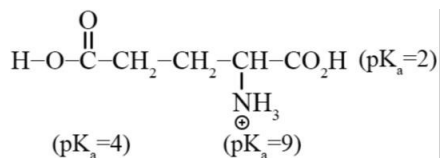


If $\text{R} = -\text{CH}_2\text{C}_6\text{H}_5$

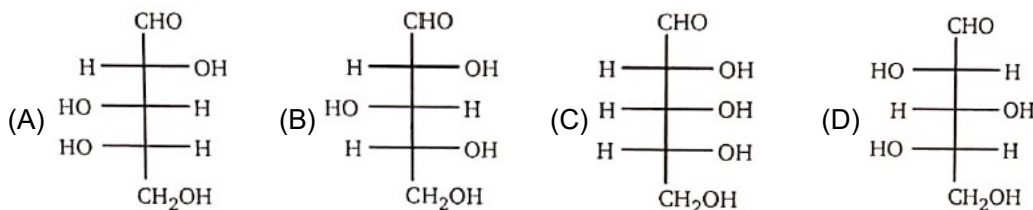
Then it is phenylalanine (Phe) and if $\text{R} = \text{CH}_3$ then it is alanine (Ala). Find the sequence of reagents from those given below to synthesize Phe – Ala.



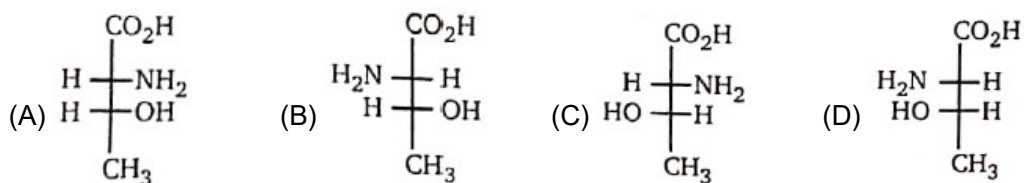
- (A) 1 and 2 (B) 1 and 4 (C) 2 and 3 (D) 3 and 4
5. At iso-electric point:
- (A) Conc. of cation is equal to conc. of anion
(B) Net charge is zero.
(C) Maximum conc. of di-polar ion (Zwitter ion) will be present
(D) All of the above
6. Which of following amino acid has lowest iso-electric point?
- (A) Glycine (B) Alanine (C) Aspartic acid (D) Lysine
7. Find iso-electric point of the given amino acid



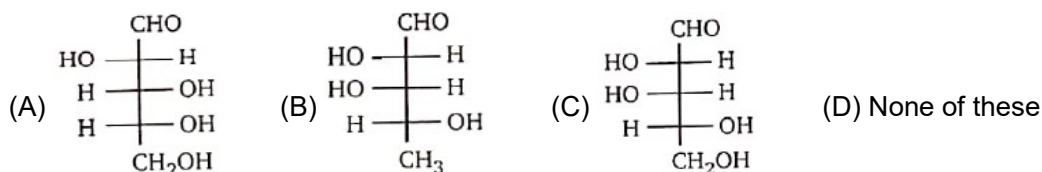
- (A) 5.5 (B) 6.5 (C) 3 (D) 5
8. Product (B) of given reaction is:
- $$\text{H}-\text{C}\equiv\text{C}-\text{H} \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4} (\text{A}) \xrightarrow[(2)\text{H}_3\text{O}^+]{(1)\text{NH}_3+\text{HCN}} (\text{B})$$
- (A) Glycine (B) Alanine (C) valine (D) Leucine
9. Which L-sugar on oxidation gives an optically active dibasic acid (2 COOH groups) ?



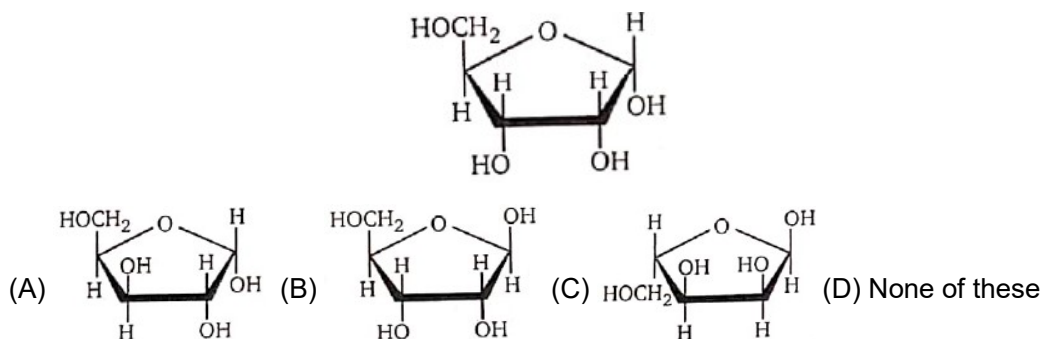
10. Which of the following is the Fischer projection of L-threonine (also known as (2S, 3R)-2-amino-3-hydroxybutanoic acid)?



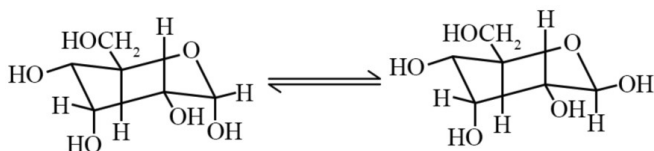
11. Among the three compounds shown below, two yield the same product on reaction with warm HNO_3 . The exception is:



12. Which of the following represents the anomer of the compound shown ?

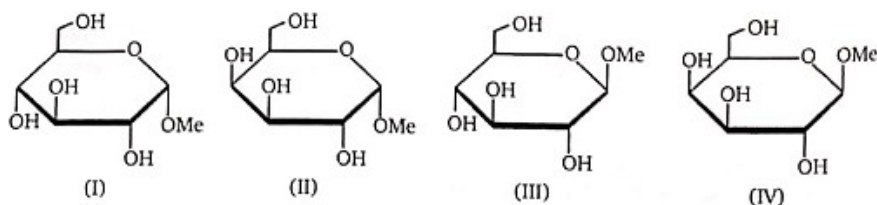


13. Which one of the statements concerning the equilibrium shown is true ?

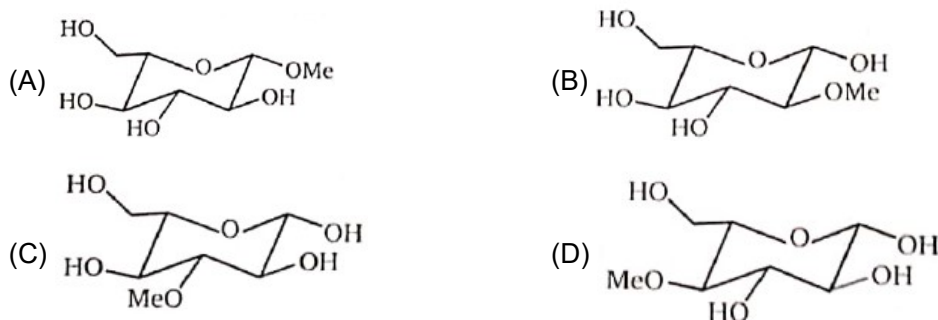
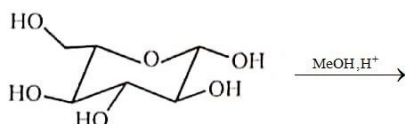


- (A) The two structures are enantiomers of each other. They have equal but opposite optical rotations and racemize slowly at room temperature
- (B) The two structures are enantiomers of each other. They racemize too rapidly at room temperature for their optical rotations to be measured
- (C) The two structures are diastereomers of each other. Their interconversion is called mutarotation
- (D) The two structures are diastereomers of each other. Their interconversion does not require breaking and making bonds, only a change in conformation

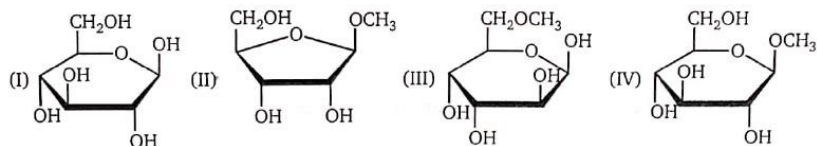
14. Identify the correct set of stereochemical relationships amongst the following monosaccharides I-IV



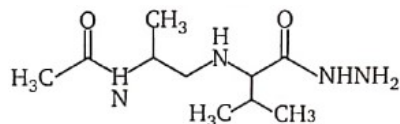
- (A) I and II are anomers ; III and IV are epimers
 (B) I and III are epimers ; II and IV are anomers
 (C) I and II are epimers ; III and IV are anomers
 (D) I and III are anomers ; I and II are epimers
15. Predict the product of the following reaction.



16. Which of the following carbohydrate(s) would not undergo mutarotation in aqueous solution?



- (A) II only (B) I, III and IV only (C) II and IV only (D) I and III only
17. The number of peptide bonds in the compound.



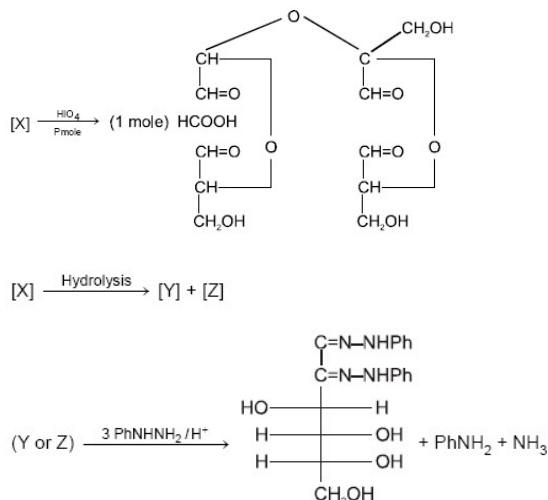
- (A) 1 (B) 2 (C) 3 (D) 4

18. **Statement 1:** The percentage of β -D(+)-glucopyranose is more than α -D(+)-glucopyranose at equilibrium.

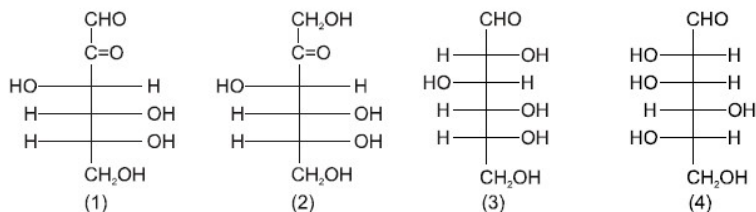
Statement 2: The presence of bulky groups in axial position makes β -D(+)-glucopyranose more stable.

- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True.
19. **Statement-1:** A solution of sucrose in water is dextrorotatory but on hydrolysis in presence of H^+ , the solution becomes levorotatory
- Statement-2:** Inversion of sugar follows first order kinetics.
- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True.

Comprehension # 1 (for Ques. 20 to 21)



20. Compounds Y and Z can be :



- (A) 1 only (B) 2, 3 (C) 1, 4 (D) 2, 3, 4

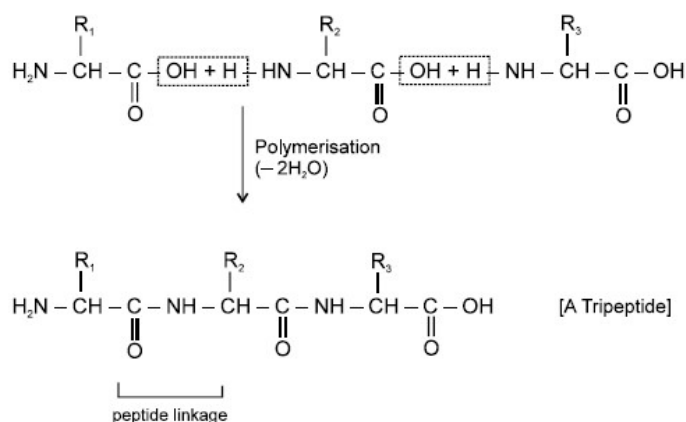
21. Number of moles (P) of HIO_4 used per moles of compound X is :

- (A) 2 (B) 3 (C) 4 (D) 5

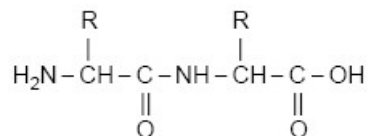
Comprehension # 2 (for Ques. 22 to 23)

Proteins are biomolecules composed of α -amino acids. An α -amino acid has a general formula $\text{R}-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$. The amino acids polymerise and form an amide linkage (peptide

linkage) between two monomeric amino acid units. The polymerisation takes place as follows



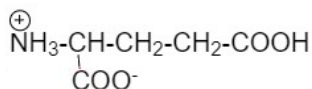
Two or more similar amino acids can also polymerise, for example a dimer will be like



22. In the above trimer, if $\text{R}_1 = \text{H}$; $\text{R}_2 = \text{CH}_3$ & $\text{R}_3 = \text{Ph}$ then total number of optically active stereoisomers will be :
 (A) 8 (B) 6 (C) 4 (D) 2
23. Which statement is incorrect about the given trimer.
 (A) it will liberate CO_2 with NaHCO_3 .
 (B) It will liberate N_2 with $\text{NaNO}_2 / \text{HCl}$
 (C) It will give yellow precipitate with 2, 4-Dinitrophenylhydrazine
 (D) It will rotate plane polarized light.

NUMERIC ANSWER TYPE

24. How many acidic group is present in given amino acid ?

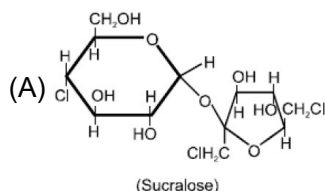


MATRIX MATCH TYPE

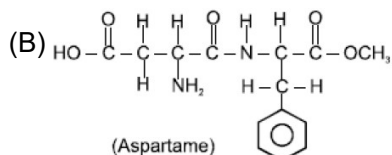
25. Match the Following:

Column 1

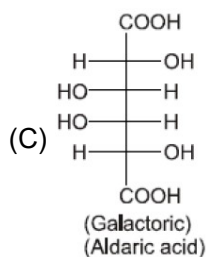
(Artificial sweeteners)



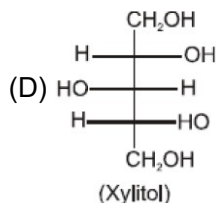
(Sucralose)



(Aspartame)



(Galactaric)
(Aldaric acid)



(Xylitol)

Column II

(Characteristics)

(p) A derivative of dipeptide

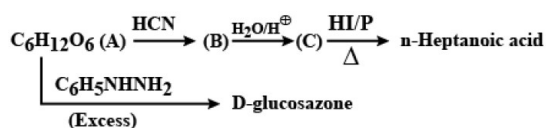
(q) A derivative of disaccharide

(r) Reduction product of an aldopentose

(s) Oxidation product of aldohexose

SUBJECTIVE ANSWER TYPE

26. Compound (A), $C_6H_{12}O_6$, is oxidized by bromine water into a monobasic acid and also reduces Tollen's reagent and Fehling's solution. It also responds to the following reactions. Identify the compound (A).



Answer Key CAPS-25

- | | | | | |
|--|---------|---------|---------|---------|
| 1. (C) | 2. (B) | 3. (D) | 4. (C) | 5. (B) |
| 6. (C) | 7. (C) | 8. (B) | 9. (A) | 10. (B) |
| 11. (B) | 12. (B) | 13. (C) | 14. (D) | 15. (A) |
| 16. (C) | 17. (A) | 18. (C) | 19. (B) | 20. (B) |
| 21. (B) | 22. (C) | 23. (C) | 24. (2) | |
| 25. $(A \rightarrow q, B \rightarrow p, C \rightarrow s, D \rightarrow r)$ | | | | |